

COMSATS University Islamabad, Vehari Campus

Information Communication Technology (CSC101)

Lab 04

Topic: MS ACCESS

E-mail: mariamsherwani434@gmail.com

Objective

This lab introduces Microsoft Access, a database management system (DBMS) that combines the relational Microsoft Jet Database Engine with a graphical user interface (GUI) and software-development tools.

Learning Outcomes

Students will gain hands-on practice with:

- Creating a new database.
- Creating and designing tables.
- Inserting, retrieving, and updating data.
- Creating forms for data entry and management.

Basic Concept of MS ACCESS

Microsoft Access is a DBMS included in professional editions of Microsoft Office. It stores data in its own format based on the Jet Database Engine and allows linking related information easily, such as customer and order data.

Access 2013 and later versions include connectivity features allowing:

- Importing or linking to data stored in other applications and databases.
- Working with SQL databases, desktop applications, servers, minicomputers, mainframes, and web servers.
- Understanding and using a wide variety of data formats.
- Exporting and importing data from spreadsheets, word processing files, or other databases.

- Integration with databases supporting ODBC, including SQL Server, Oracle, and DB2.
 - Application development for software projects.
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Activity 1: Create a Blank Database

1. Open Microsoft Access.
2. Select **Blank desktop database**, enter a name, and click **Create**.
3. Access creates a new blank database and opens an empty table.

Activity 2: Creating Tables

1. Tables store all data in the database. Other database objects depend heavily on tables, so start by creating them first.
2. Consider all requirements and determine necessary tables before creating them.

Example — Employee Table:

Field Name	Data Type
EmployeeID	AutoNumber
FirstName	Short Text
LastName	Short Text
Address1	Short Text
Address2	Short Text
City	Short Text
State	Short Text
Zip	Short Text
Phone	Short Text
PhoneType	Short Text

Steps:

1. Open a blank database.
2. Access automatically creates an ID field as AutoNumber; rename it to **EmployeeID** using Name & Caption in the Ribbon.

3. Add other fields by selecting **Click to Add** → **Short Text** and entering field names.
 4. Save the table and enter a name (e.g., tblEmployee).
 5. In Design View, verify fields and data types side by side.
 6. Set **EmployeeID** as the primary key by selecting it and clicking **Primary Key**.
 7. Make additional changes in Datasheet View as needed (e.g., updating PStatus field).
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Lab Task 1

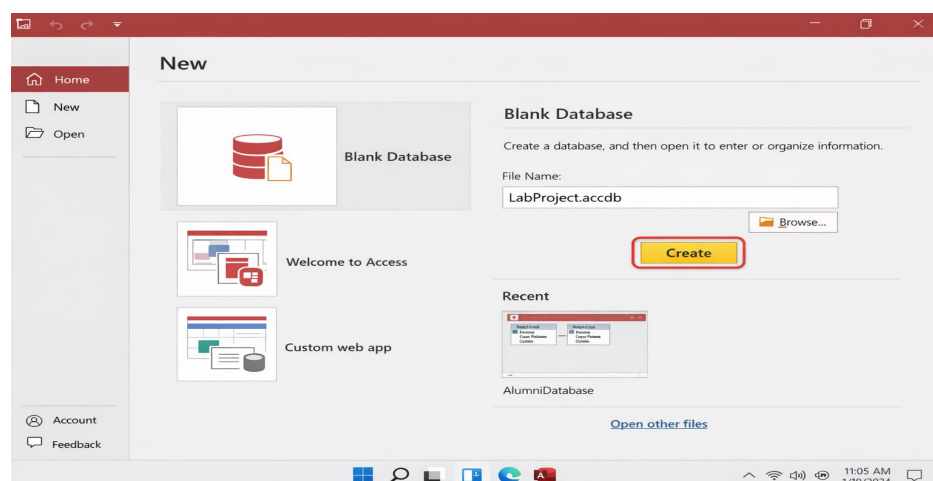
Create database of your lab project. Create tables for each activity. Apply the following queries on the tables:

- Insert data in the tables.
- Retriew data from the tables.
- Update the data.
- Create forms for all the activites.

Solution:

Step 1 — Create Database

- Open Microsoft Access.
- Select **Blank Database**.
- Enter the database name: LabProject.accdb.
- Click **Create**.



Step 2 — Create Tables (3 Tables Required)

1. Students Table

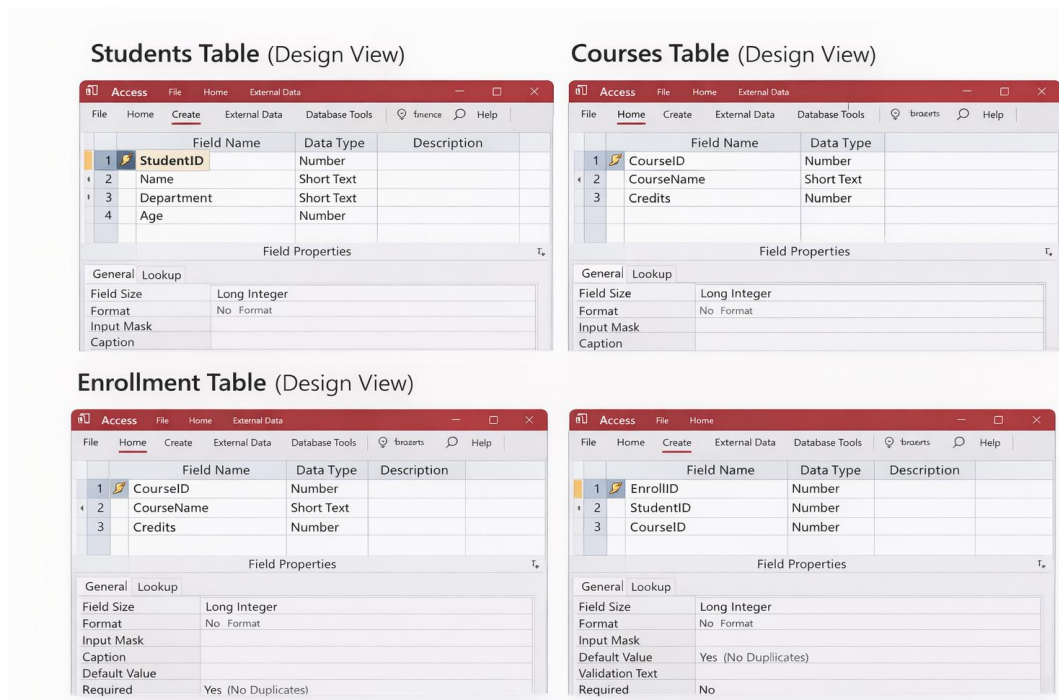
- Go to **Create → Table Design**.
- Add fields:
 1. StudentID (Number, Primary Key)
 2. Name (Short Text)
 3. Department (Short Text)
 4. Age (Number)
 5. Save table as **Students**.

2. Courses Table

- Go to **Create → Table Design**.
- Add fields:
 1. CourseID (Number, Primary Key)
 2. CourseName (Short Text)
 3. Credits (Number)
 4. Save table as **Courses**.

3. Enrollment Table

- Go to **Create → Table Design**.
- Add fields:
 1. EnrollmentID (Number, Primary Key)
 2. StudentID (Number)
 3. CourseID (Number)
 4. Save table as **Enrollment**.



Step 3 — Insert Data (Datasheet View for All Tables)

1. Students Table Data

StudentID	Name	Department	Age
1	Ali	Computer Science	20
2	Sara	IT	21

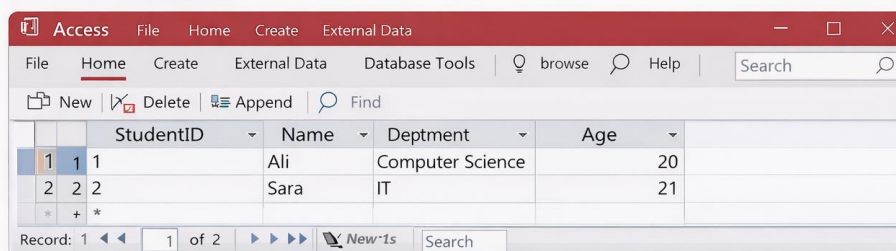
2. Courses Table Data

CourseID	CourseName	Credits
101	Database Systems	3
102	Programming	4

3. Enrollment Table Data

EnrollID	StudentID	CourseID
1	1	101
2	2	102

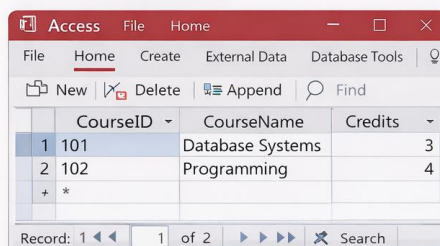
3A. Students Table Data



The screenshot shows the Microsoft Access application window with the 'Students' table open in Datasheet View. The table has four columns: StudentID, Name, Deptment, and Age. There are two records: StudentID 1 (Ali, Computer Science, 20) and StudentID 2 (Sara, IT, 21). The status bar at the bottom indicates 'Record: 1 of 2'.

StudentID	Name	Deptment	Age
1	Ali	Computer Science	20
2	Sara	IT	21

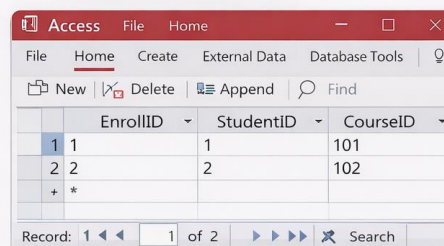
3B. Courses Table Data



The screenshot shows the Microsoft Access application window with the 'Courses' table open in Datasheet View. The table has three columns: CourseID, CourseName, and Credits. There are two records: CourseID 101 (Database Systems, 3 credits) and CourseID 102 (Programming, 4 credits). The status bar at the bottom indicates 'Record: 1 of 2'.

CourseID	CourseName	Credits
101	Database Systems	3
102	Programming	4

3C. Enrollment Table Data



The screenshot shows the Microsoft Access application window with the 'Enrollment' table open in Datasheet View. The table has three columns: EnrollID, StudentID, and CourseID. There are two records: EnrollID 1 (StudentID 1, CourseID 101) and EnrollID 2 (StudentID 2, CourseID 102). The status bar at the bottom indicates 'Record: 1 of 2'.

EnrollID	StudentID	CourseID
1	1	101
2	2	102

3C. Enrollment Table Data

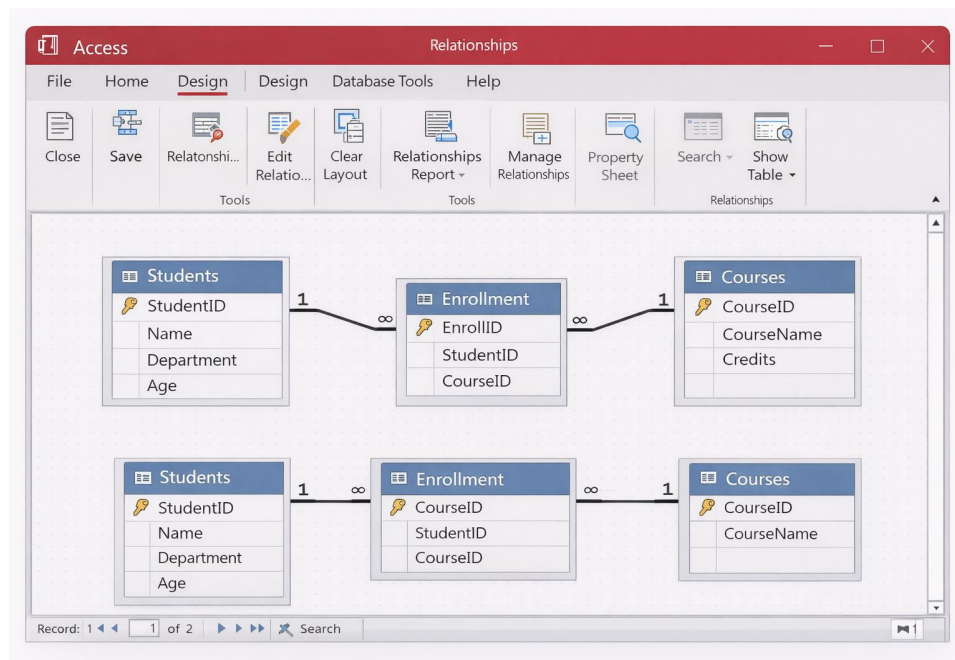


The screenshot shows the status bar of the Enrollment table, indicating 'Record: 1 of 2'.

Step 4 — Create Relationships

- Go to **Database Tools** → **Relationships**.
- Add **Students**, **Courses**, **Enrollment** tables.
- Create relationship lines:
 1. Students.StudentID → Enrollment.StudentID
 2. Courses.CourseID → Enrollment.CourseID
- Enable **Referential Integrity**.

Purpose: Ensure data consistency across tables.



Step 5 — Queries Retrieve + Update)

1. Insert Data

- Open the table in Datasheet View.
- Type new rows directly or create an **Append Query**.

2. Retrieve Data Query

- Go to **Create** → **Query Design**.
- Add **Students**, **Courses**, **Enrollment** tables.
- Join fields:
 1. Students.StudentID = Enrollment.StudentID
 2. Courses.CourseID = Enrollment.CourseID
- Select fields:
 1. Students.Name
 2. Courses.CourseName
- Run query to view results -> Click **Run (!)**.

Query Result Output

Name	CourseName
Ali	Database Systems
Sara	Programming

3. Update Query

- Go to **Create** → **Query Design**, add the table.
- Click **Update** in the Ribbon.
- Select **Students** table.
- Drag the field to update (e.g., Age) to the grid.
- In **Update To**, type the new value (e.g., 22).
- Add criteria (e.g., StudentID = 1) and run the query.
- Run query to update the data.

5A — Query Design (Retrieve Data)

5B — Query Result Output

Name	CourseName
Ali	Database Systems
Sara	Programming

5C — Update Query

Query1

UPDATE Students

Field:	Age	Table:	Students	Update To:	22	Criteria:	1
Table:	Students	Students					
Update To:							
Criteria:							
Or:							

Step 6 — Forms for All Activities

1. Student Entry Form

- Go to **Create → Form**.
- Select **Students** table.
- Form fields: StudentID, Name, Department, Age
- Save as **StudentForm**.

2. Course Entry Form

- Go to **Create → Form**.
- Select **Courses** table.
- Form fields: CourseID, CourseName, Credits
- Save as **CourseForm**.

3. Enrollment Form

- Go to **Create → Form**.
- Select **Enrollment** table.
- Form fields: EnrollID, StudentID, CourseID
- Save as **EnrollmentForm**.

The image displays three screenshots of Microsoft Access forms, each titled with a number and a description:

- 6A — Student Entry Form:** This form is titled "StudentForm". It contains four input fields: "Student ID" with the value "1", "Name" with the value "Ali", "Department" with a dropdown menu showing "Computer Science", and "Age" with the value "22". At the bottom, there are navigation buttons (back, forward, etc.) and "Save" and "Delete" buttons.
- 6B — Course Entry Form:** This form is titled "CourseForm". It contains three input fields: "Course ID" with the value "1", "Course Name" with the value "Database Systems", and "Credits" with the value "3". At the bottom, there are navigation buttons and "Save" and "Delete" buttons.
- 6C — Enrollment Form:** This form is titled "EnrollmentForm". It contains four input fields: "Enroll ID" with the value "1", "Student ID" with a dropdown menu showing "1", "Course ID" with a dropdown menu showing "1", and "Student ID" with the value "1". At the bottom, there are navigation buttons and "Save" and "Delete" buttons.